

For this homework, you shall use the [ERDPlus](#) website to create your Entity Relationship Diagrams (ERDs).

Provide an ERD for each of the problem descriptions below.

Where applicable use the Vet Clinic schema from the **Instructor_File: cpts451hw01rds.zip**

- Q1.** It is possible for a [Veterinarian](#) (i.e., Vet) to see many different patient animals in the clinic on a given day. Any patient animal may be seen by different Vets over its lifetime. Each time a Vet sees an animal it is called a “visit”. A visit can be identified by the date upon which it occurred. No animal is seen by the same Vet within the same day. It is possible that a Vet may see the same animal more than once in that animal’s lifetime. A Veterinarian may be singularly identified by their very own Vet Clinic employee identifier. Other veterinarian information that may be found in the Vet Clinic database would be their first & last name, their address (several attributes) and a phone number.
- Q2.** The Vet Clinic stocks many medications. Each medication has a unique National Drug Code (NDC) identifier [FDA format: [21 CFR 207.33 \(b\)](#)]. A medication has a common name, a manufacturer, the provided form (e.g., *liquid, pill, cream, etc.*), its dosage amount, number/amount per package and number of packages currently available from the inventory stock. A patient animal of a Vet Clinic may be prescribed none or some medications in form of a prescription. An individual prescription is only ever for a single animal patient. A single animal patient may accrue many prescriptions over its lifetime. It is possible that a medication may appear in more than one prescription (*assuming the stock is available*). Each prescription has a unique electronically assigned identifier on it, as well as a date, and the Veterinarian making out the prescription.
- Q3.** The Vet Clinic has decided that within an animal patient entity, the [Species](#) attribute is repetitious should be its own separate lookup list. Revamp the animal patient entity so that it now uses a lookup list for the species. An animal patient is always a species but is never more than one species. In the species lookup list, there is a unique species identifier, the common name, and the scientific name (e.g., [European hamster](#) and *cricetus cricetus*). A species may be listed without an associated Vet Clinic patient. The Vet Clinic may have more than one animal patient of the same species.
- Q4.** A [vehicle repair shop](#) has a database, within that are vehicle things. A vehicle has a unique license number (*state issued*), a make, a model, and the year it was manufactured. The shop repairs many kinds of vehicles, such as motorcycles, automobiles, trucks, and boats. For trucks the number of axles and load capacity is kept track of. For motorcycles, we need to if it is 2 or 3 wheels and if it has an electric or gas engine. For automobiles, data is retained regarding its number of engine cylinders and fuel type. For boats we need to know how many engines it has, whether it is an open or closed (cabin?) style, and its displacement. A vehicle shall always be a kind of vehicle, but never more than one kind.
- Q5.** A vehicle repair shop has a database, within that is a bunch of [standardized repair tasks](#). When a vehicle has repair tasks assigned to it, it is possible that some tasks must be done in a particular order. Given that, each task may have an identified prerequisite task that should be performed first. It is also possible that a task may not be a prerequisite for any other task. All repair tasks are given a unique identifier, a title/description, and a vehicle repair procedure reference (e.g., *PROC-AUTO-027 Rev 3*).
- Q6.** A [wedding planner](#) has many events. She assigns each event a unique identifier, and that event also has a known location, date, time, duration, and associated client number. Associated with that event may be many different types of vendors, such as: Caterer, Florist, Photographer, Artists: Hair, Makeup & Nails, Cake Baker, Jeweler, Attire Professionals, etc. Each Vendor has a unique identifier, a business name, an owner name, a phone number, billing address (*several fields*), and a type. It is possible, the event is small enough that the Bride & Groom just take of all of that themselves, and don’t really need any vendors.

- Q7.** Your cell phone has a [Contacts](#) database. Each contact has a first & last name, a primary phone number (*also a possible alternate phone number*) and may have a relationship with another contact (*e.g., child, parent, significant other, etc.*). Each contact may have additional information such as an address, birthday, and maybe an email.
- Q8.** The [Vidiians](#) need a database, within it they are tracking all indigenous Blungils. So far, each blungil has a special identifying Dinklar feature set. Key blungil information includes their ynshu cycle (*number of occurrences per Vidiia year, if any*), the total number of tentacularian appendages, inglomeration temperature (*centigrade*), and last known location (*Terran GPS standard [format](#)*), additionally a possible sobriquet (*unique*). The Vidiians may keep a blungil as a pet, as livestock to harvest their lactations (*medicinal purposes*), and/or as a meat/food source, it rumored that wild blungils still exist and roam the Ogygis Undae desert. The general perambulation rate of a blungil has never been fully defined, but it still may need hoof coverings. Given this a hoof covering is a thing with a particular stock/inventory identifier, a blungil hoof size value (*0-67 with a single decimal*), [Zloty](#) price, its ambient light reflectivity value data set (*e.g., hue, saturation, luminosity*) and kleptar ratio. It is possible that a blungil may require different hoof coverings depending upon the size of their various tentacularian appendages.
- Q9.** The [Alliance of Therapy Dogs](#) (*nonprofit*) needs a database. They have Sponsors and want to keep track of their charitable donations. The sponsor information includes that organizations formal name, their website URL, Point-of-Contact person name and phone number. The donation list includes the date it was made, the amount (*in Euros or US Dollars*), the sponsor who made it, and their donations are numbered per each sponsor starting at one for the first donation and incrementing by one for subsequent donations.
- Q10.** The [International Council of Beverages Associations](#) has decided their USA affiliated organization, the [American Beverage Association](#) (*due to the [FDA's nutrition labeling regulations](#)*), that they now wish to keep track of their published scientific papers produced by their research division. A specific paper's bibliographic information may include: author(s) name (*e.g., one or several*), publication title, article title (*if part of a larger anthology*), publication date, publication place, publisher, volume number of a journal, magazine or encyclopedia (*where applicable*), and the starting & ending page number(s). The Author list is separate and contains first and last names, honorific title (*prefix*), highest level academic degree type, title & date bestowed, an email, a phone number (*international format*), and whether they are for/against recycling of beverage containers (*e.g., aluminum cans and/or plastic bottles*).

Homework Notes:

Please format your answers in an aesthetically pleasing manner. Whatever you decide, be consistent.

Annotate each ERD with a problem identifier and your initials such as: **Q1-[abc](#)**

Export the individual ERDs as PNG image files from the web software. Name the individual picture files appropriately, mainly so I don't have to guess which problem it is for and who did it.

Submit a single ZIP file containing all your answers to the [Homework 02 Drop Box](#) on the class website. Name your file: **cpts451hw02[abc](#).zip** (*where the [abc](#) part is your initials*).

You may include an additional **ReadMe.txt** as needed with any additional explanatory comments that you feel should be considered as a part of the assignment review process.

I shall be acting in the "Client" role to answer any assignment content questions possibly needing clarification, up until three days before the due date.